

Lineage-Primary Preservation and the Reordering of Authenticity

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1. Introduction

Conventional digital preservation defines success by the maintenance of exact bit-level fidelity, where any deviation detected via checksums is treated as failure. This chapter argues that such an assumption is untenable over long temporal horizons characterized by format obsolescence and synthetic media production. Instead, a lineage-primary preservation model is proposed, where identity continuity is carried through documented transformation histories. Bit-level preservation is retained only as a non-authoritative historical reference rather than the final arbiter of authenticity.

2. Limits of Bit-Level Preservation Over Millennial Timescales

While bit-level preservation maximizes fidelity under ideal conditions, it is statistically unlikely to survive centuries due to organizational and cultural risks, such as gaps in stewardship or loss of interpretive environments. When these systems fail, they often do so catastrophically, rendering files technically intact but uninterpretable. Bit-level approaches offer limited resilience against the partial and contingent degradation that occurs over millennial horizons.

3. Lineage as the Primary Carrier of Identity

A lineage-primary model treats individual files as transient carriers rather than the preserved object itself. Success is evaluated by whether successive instantiations remain recognisably continuous with an originating identity. Every meaningful transformation—such as migration, repair, or reconstruction—is recorded as a “lineage event” in a provenance graph. This approach accepts that earlier states may be irrecoverable in their exact form, but their transformation remains legible.

4. Species Descriptors and Bounded Drift

To prevent uncontrolled interpretive drift, lineage is constrained by a minimal species descriptor. This descriptor defines:

- The origin anchor of the work.
- Its semantic intent.
- Its structural grammar.
- The bounds of permissible variation.

Deviation within these bounds is considered acceptable drift, while deviation beyond them terminates that lineage branch.

5. Repositioning Fixity as Reference Rather Than Authority

In this system, a bit-checked reference copy is preserved as an immutable, evidentiary, and non-authoritative artifact. It functions like a “fossil record” for calibration. If the reference is lost, preservation does not fail because identity resides in the lineage rather than any single carrier.

6. Implications for Synthetic Media and Artificial Intelligence

In a lineage-primary system, visual accuracy alone cannot support authenticity claims. Synthetic artifacts lacking documented ancestry cannot assert continuity with historical identities. AI-assisted restorations are

recorded as lineage events, making AI a tool within the chain rather than a mechanism capable of fabricating an origin.

7. Comparative Durability Over Long Horizons

Over a 1,000-year horizon, a lineage-primary system is more likely to yield a recognizable artifact than a fixity-only system. It allows for “graceful degradation” rather than “cliff-edge failure”.

8. Conclusion

Preservation is reframed not as the indefinite freezing of data, but as the sustained legibility of identity through time. By reordering the hierarchy, lineage carries identity forward while bit-level preservation serves as historical evidence.